Math 3
CC 6

Name: Arjun Rakesh
Date: _____ Per: 6

Directions: Complete each of the following problems. Show work for all problems involving calculations. All answers must be in simplest form.

III) Rational and Irrational Numbers (Try 4)

1) Which of the following are rational numbers? **Circle all that apply.** (2 pts)

-1/2

(A) 2.879458134 (Terminating)
(B) $1.\overline{345}$
(C) $\frac{17}{8}$
(D) $\sqrt{64}$
(E) $\sqrt[3]{64}$
(F) $\sqrt[3]{25}$

2) Estimate $\sqrt{140}$ to the nearest tenths place. (1 pt)

$\sqrt{121} < \sqrt{140} < \sqrt{144}$

11 $\frac{19}{23}$ 12

144 − 121 = 23

≈ 11.8

3) **USING THE METHOD SHOWN IN CLASS.** Write $0.0\overline{7}$ as a fraction. (The line is ONLY over the seven). (2 pts)

$100x = 7.7777$
$x = 0.0777$
$99x = 7.7$
$\frac{99x}{99} = \frac{7.7 \times 10}{99 \times 10}$
$\frac{990x}{990} = \frac{77}{990}$
$\frac{77}{990} = \frac{7}{90}$

IV) Exponent Properties (Try 2)

For problems 4-6, simplify the expression. Write your final answer using only positive exponents.

4) $1{,}456{,}789{,}874{,}500^0$ (½ pt)

$= 1$

5) $(x^5)^4 \cdot x^{12}$ (1 ½ pts)

-1

$(x^{5+4}) \cdot x^{12}$ (multiply)
$1x^{32}$
$(x^9) \cdot x^{12}$
x^{21}

6) $\frac{a^2 b^{10}}{a^4 b^{-60}}$ (1 ½ pts)

-1

$\left(\frac{a^2}{a^4}\right)\left(\frac{b^{10}}{b^{-60}}\right)$
$\frac{b^{70}}{a^2}$
$(a^{2-4}) \times (b^{10+60})$
$\frac{1}{a^2} \times b^{70}$

7) If you triple 3^{99} what is your result? (½ pt)

3^{100} because $3^1 \times 3^{99} = 3^{100}$
(3 is the same as 3^1)

8) Solve for x. Reminder to show work. (1 pt)

-1

$(10^{12})^x = 10^{20}$ (Multiply)
$x = 8$
$(10^{12+x}) = 10^{20}$
$10^{20-12} = x$
$10^{20-12} = 8$

Overall Score For This Standard: 4L

Overall Score For This Standard: 2

V) Scientific Notation (Try 1)

9) Convert 1.89×10^{-5} to standard form. (1 pt)

0.0000189

0.0000189

10) Is 92.4×10^{9} in scientific notation? Explain why or why not. (1 pt)

His not because scientific notation requires the non-exponent number to be less than 10 and greater than 0

Ex: 9.24×10^{10}

11) Simplify the expression, write your final answer in scientific notation. (1 ½ pts)

$(9.9 \times 10^{42}) + (8 \times 10^{41})$

$(99+8) \times 10^{41}$

$(107) \times 10^{41}$

$107 \times 10^{41} = 1.07 \times 10^{43}$

~~12)~~ -½ Eziah consumed 8.4×10^{12} burritos in 4×10^{2} days. Assuming he ate the same amount of burritos per day, how many burritos did he consume in one day during this time? (1 ½ pts)

$4\overline{)8.4}$ → 2.1, −8, 04

$\frac{8.4 \times 10^{12}}{4 \times 10^{2}}$

$\left(\frac{8.4}{4}\right) \times \left(\frac{10^{12}}{10^{2}}\right)$

$2.1 \times 10^{12-2}$

2.1×10^{10} BURRITOS

Overall Score For This Standard: 4½

Complete the drawing! It's not a cat, it's a...

